

Student Performance Data Warehouse and OLAP Analysis

Data Warehouse, OLAP Cube and Analytical Report

Dataset: Student Performance (GCE AL Exam_2020 - Sri Lanka)

Source: Kaggle dataset by Sasika Amarasinghe

Technology Used

Component	Technology
Data Processing / ETL	Python / Pandas / Google Colab
Data Warehouse	PostgreSQL
Database Management	pgAdmin
OLAP Analysis	Python / Pandas
Visualizations	Matplotlib

This report documents the complete six-step OLAP development process, from business requirements and data source identification to cube construction, analytical operations, visualization and validation.

1. Define the Business Requirements

The purpose of the OLAP solution is to analyse Sri Lankan GCE Advanced Level student performance and identify performance patterns across academic streams, subjects, gender groups and syllabuses.

Main Business Question

Which subjects have the highest average Z-score within each academic stream, and how does performance differ by gender and syllabus?

Supporting Analytical Questions

- Which academic stream has the highest average Z-score?
- Which subjects have the highest average Z-scores?
- How does student performance differ between female and male students?
- Does average performance differ between the new and old syllabuses?
- How many students are represented in each academic stream?
- What are the average district and island ranks?

Relevant Dimensions

Dimension	Purpose
Student	Student identity, gender and birth date
Subject	Subject-level performance analysis
Stream	Academic stream comparison
Exam	Exam year analysis
Syllabus	New versus old syllabus comparison

2. Identify the Data Source

The source dataset is **Student Performance (GCE AL Exam_2020 - Sri Lanka)**, obtained from Kaggle dataset by Sasika Amarasinghe. The dataset contains student-level GCE Advanced Level examination information.

Original Dataset

Property	Value
Original rows	337,553
Original columns	19
Examination year	2020
Subjects per student	3

ETL / Data Transformation

- Text values were cleaned and stripped.
- Z-score, district rank and island rank were converted to appropriate numeric types.
- Birth day, month and year were combined into a birth_date field.
- Subject-specific columns were transformed into a subject-performance structure.
- Missing values were retained where they represented missing source information.
- The transformed subject-level dataset contains 1,012,659 fact records.

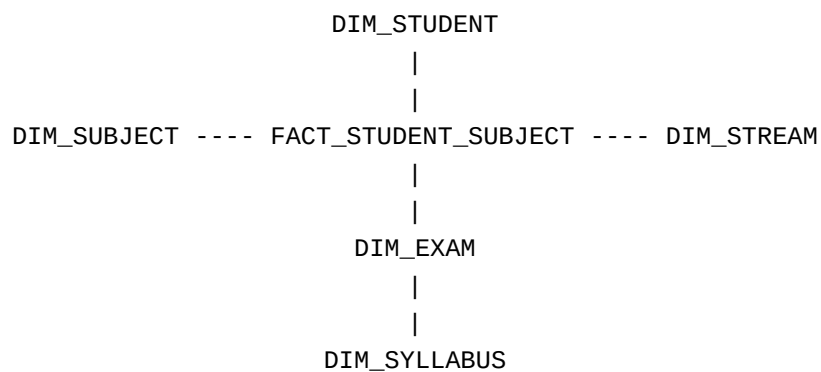
3. Define Dimensions and Hierarchies

A star schema was designed for the data warehouse. The central fact table stores student-subject performance, while dimension tables provide the descriptive context required for OLAP analysis.

Dimension	Hierarchy / Level	Records
Student	Student → Gender	337,553
Subject	Subject	62
Stream	Stream	8
Exam	Year	1
Syllabus	Syllabus	2

Star Schema

The fact_student_subject table is the central fact table. It is connected to dim_student, dim_subject, dim_stream, dim_exam and dim_syllabus through foreign keys.



4. Choose Facts / Measures

The grain of the fact table is **one row per student per subject performance**. Quantitative attributes were selected as measures for analytical calculations.

Measure	Aggregation	Purpose
Z-score	Average	Main performance indicator
District Rank	Average	Compare district-level ranking
Island Rank	Average	Compare national/island ranking
Student Count	Distinct Count	Measure participation
Grade	Count	Analyse grade distribution

Important Measure Handling

Z-score is the main numerical performance measure and is aggregated using the average. District Rank and Island Rank can also be averaged for comparative analysis. Student Count is calculated using distinct student keys. Grade is categorical rather than numerical, therefore grade values are counted instead of averaged.

The overall average Z-score in the fact data is **0.3054**, with a minimum of **-1.9383** and a maximum of **3.5583**.

5. Build and Populate the OLAP Cube

The PostgreSQL star schema was loaded into Google Colab using Pandas. The fact table was joined with all five dimension tables to create an analytical dataset.

Cube Construction

```
cube_data =  
fact_student_subject  
+ dim_student  
+ dim_subject  
+ dim_stream  
+ dim_exam  
+ dim_syllabus
```

```
OLAP cube:  
index = Stream + Subject  
columns = Gender + Syllabus  
measure = Average Z-score
```

The resulting Pandas OLAP cube contains 215 rows and 6 gender/syllabus combinations. Missing combinations are represented as NaN values because there are no records for those particular dimension combinations.

OLAP Operations Implemented

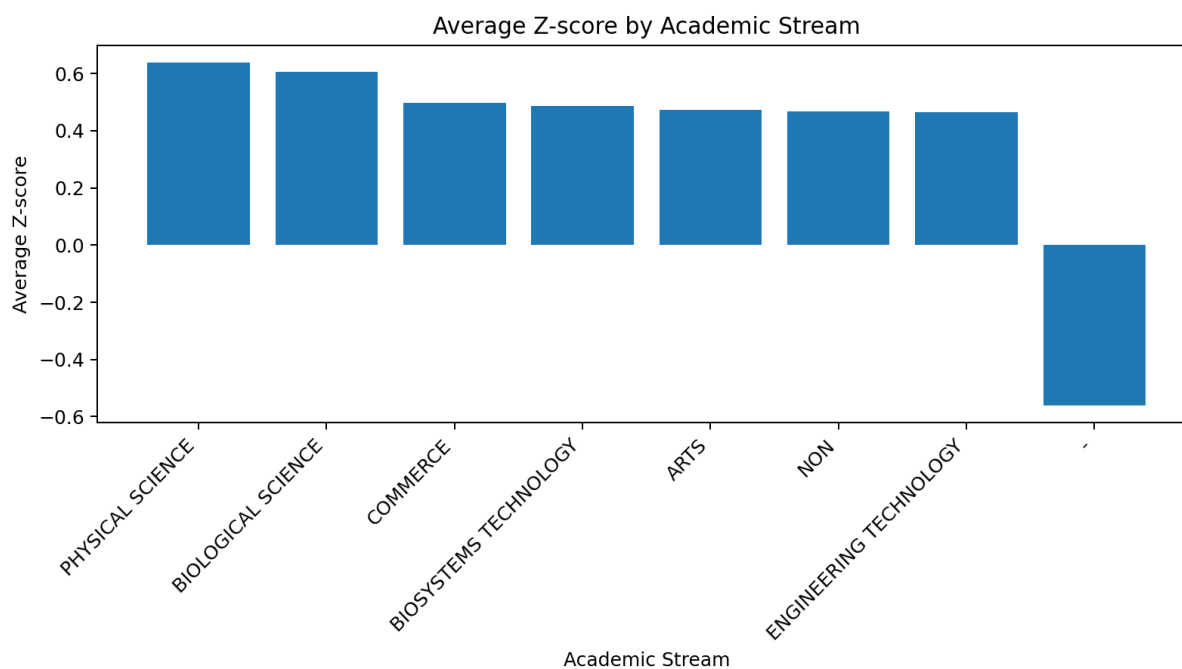
Operation	Implementation
Roll-up	Stream-level performance
Drill-down	Stream → Subject performance
Slice	Physical Science stream
Dice	Physical + Biological Science, Female + Male, New + Old

6. OLAP Analysis Results

6.1 Roll-up: Average Z-score by Academic Stream

Roll-up summarizes detailed student-subject records at the academic stream level.

Stream	Average Z-score	Student Count
PHYSICAL SCIENCE	0.638	21,776
BIOLOGICAL SCIENCE	0.606	25,768
COMMERCE	0.497	39,877
BIOSYSTEMS TECHNOLOGY	0.487	7,833
ARTS	0.471	76,547
NON	0.466	2,740
ENGINEERING TECHNOLOGY	0.464	12,498
-	-0.560	150,514



The highest average Z-score among the listed streams is observed for **PHYSICAL SCIENCE** with an average of **0.6380**.

6.2 Drill-down: Stream to Subject

Drill-down moves from a higher-level stream view to subject-level performance. This allows detailed investigation of subjects within each stream.

Stream	Subject	Average Z-score	Students
-	ENGLISH	-0.146	1,054
-	FRENCH	-0.200	626
-	MATHEMATICS	-0.214	434
-	JAPANESE	-0.302	946
-	BIO-RESOURCE TECHNOLOGY	-0.326	22
-	HISTORY OF MODERN WORLD	-0.355	34
-	GERMAN	-0.370	180
-	BUSINESS STATISTICS	-0.376	375
-	HINDI	-0.384	196
-	HISTORY OF SRI LANKA & MODERN WORLD	-0.408	124
-	COMBINED MATHEMATICS	-0.413	20,245
-	GREEK & ROMAN CIVILIZATION	-0.417	135
-	BIOLOGY	-0.420	25,164
-	PHYSICS	-0.430	41,285
-	CHEMISTRY	-0.433	42,381
-	FOOD TECHNOLOGY	-0.442	105
-	AGRICULTURAL SCIENCE	-0.462	13,641
-	INFORMATION & COMMUNICATION TECHNOLOGY	-0.465	19,325
-	CIVIL TECHNOLOGY	-0.480	194
-	AGRO TECHNOLOGY	-0.482	732

The detailed drill-down result contains **218** stream-subject combinations.

6.3 Slice: Physical Science

A slice selects one value from a dimension. In this analysis, the cube was sliced to the **PHYSICAL SCIENCE** stream.

Subject	Average Z-score	Student Count
CHEMISTRY	0.638	21,776
COMBINED MATHEMATICS	0.638	21,776
PHYSICS	0.638	21,776

The three Physical Science subjects have the same average Z-score in this dataset. This reflects the structure of the source data and has not been artificially changed.

6.4 Dice: Multiple Dimension Filters

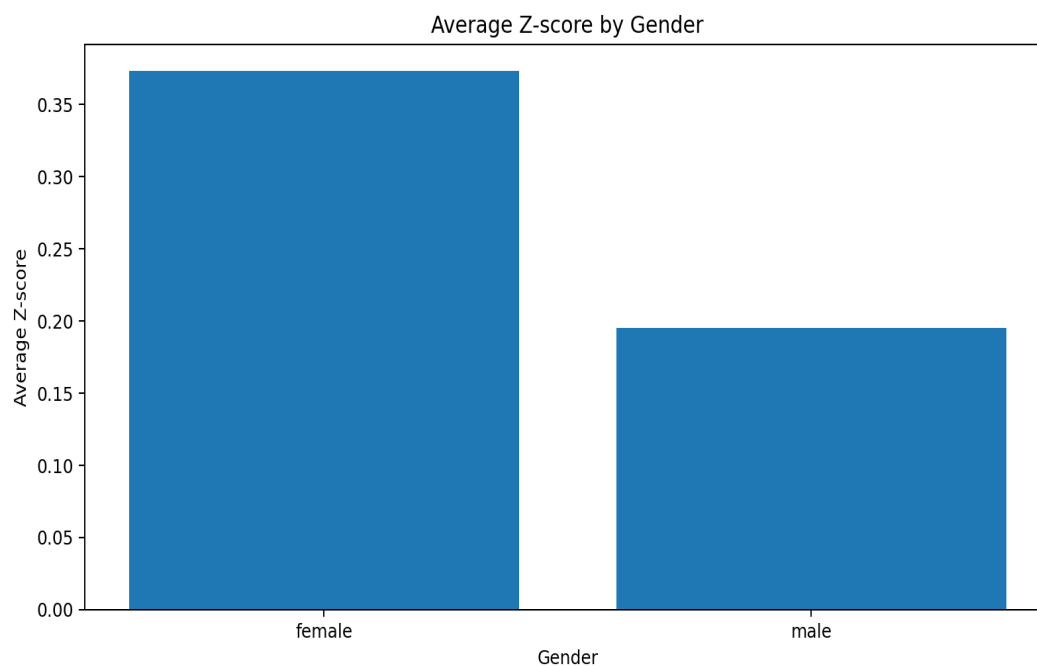
The dice operation selects multiple values across several dimensions. The analysis uses Physical Science and Biological Science streams, female and male genders, and new and old syllabuses.

Stream	Gender	Syllabus	Average Z-score	Students
BIOLOGICAL SCIENCE	male	new	0.731	5,434
BIOLOGICAL SCIENCE	male	old	0.589	1,080
BIOLOGICAL SCIENCE	female	new	0.586	15,654
BIOLOGICAL SCIENCE	female	old	0.507	3,557
PHYSICAL SCIENCE	male	new	0.713	12,598
PHYSICAL SCIENCE	female	new	0.554	7,745
PHYSICAL SCIENCE	male	old	0.474	858
PHYSICAL SCIENCE	female	old	0.387	557

7. Gender Performance Analysis

Gender performance was analysed using average Z-score and distinct student count. The 'major error' source category was not included in the meaningful female/male comparison because it contains only 24 records and represents a data-quality category rather than a gender.

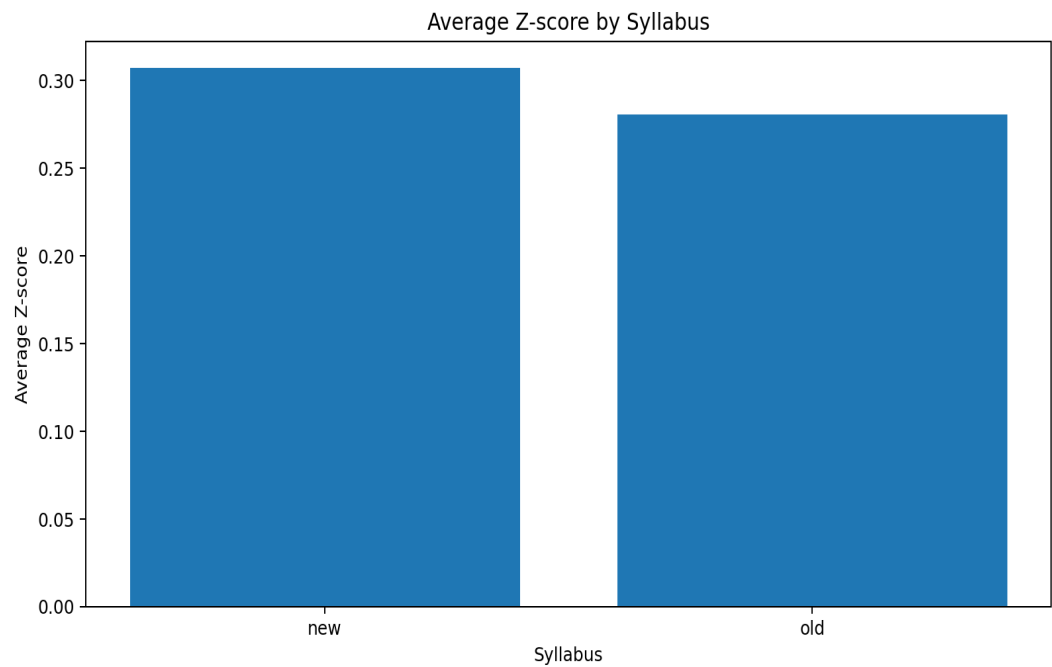
Gender	Average Z-score	Student Count
female	0.373	193,326
male	0.195	142,719



Among the female and male categories, **female** has the higher average Z-score at **0.3729**.

8. Syllabus Performance Analysis

Syllabus	Average Z-score	Student Count
new	0.307	306,211
old	0.281	31,342

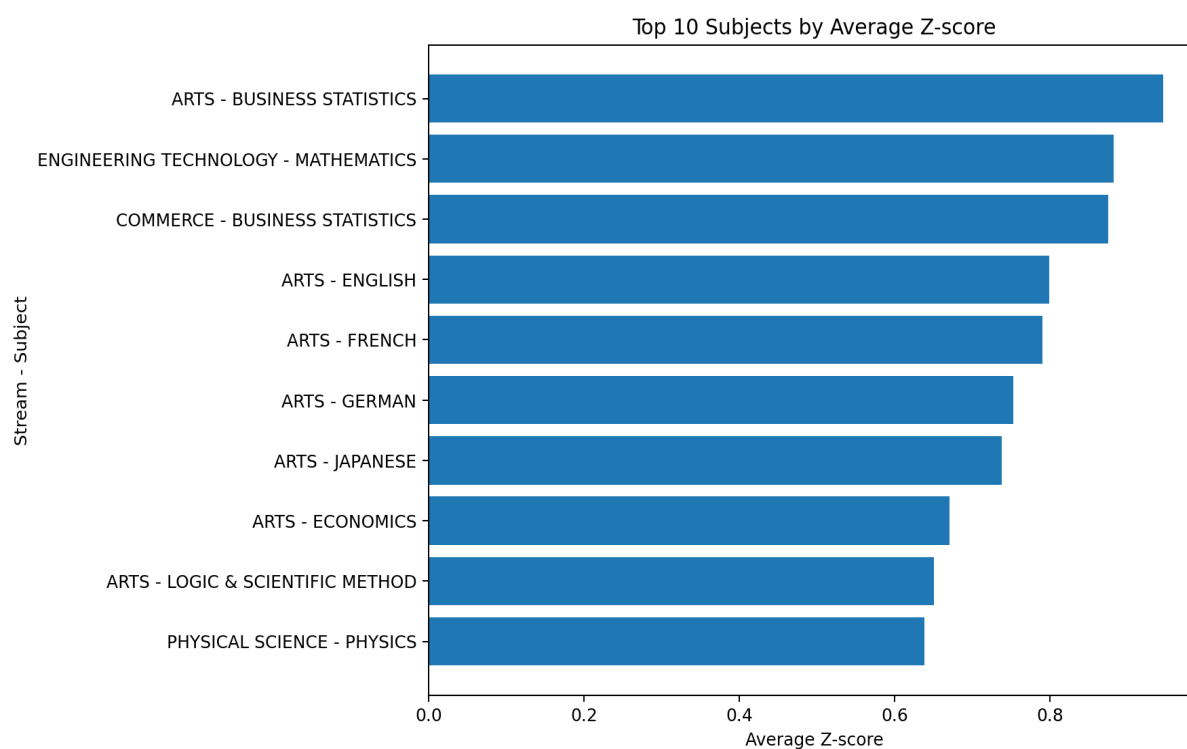


The **new** syllabus has the higher average Z-score of **0.3072**, compared with **0.2805** for the old syllabus.

9. Top Subject Performance

To avoid misleading conclusions from subjects with very small numbers of students, subjects were filtered to include at least 100 students before identifying the highest average Z-scores.

Stream	Subject	Average Z-score	Students
ARTS	BUSINESS STATISTICS	0.945	215
ENGINEERING TECHNOLOGY	MATHEMATICS	0.882	241
COMMERCE	BUSINESS STATISTICS	0.874	563
ARTS	ENGLISH	0.799	953
ARTS	FRENCH	0.790	647
ARTS	GERMAN	0.753	249
ARTS	JAPANESE	0.738	1,032
ARTS	ECONOMICS	0.670	4,788
ARTS	LOGIC & SCIENTIFIC METHOD	0.651	6,604
PHYSICAL SCIENCE	PHYSICS	0.638	21,776
PHYSICAL SCIENCE	CHEMISTRY	0.638	21,776
PHYSICAL SCIENCE	COMBINED MATHEMATICS	0.638	21,776
NON	PHYSICS	0.631	1,811
NON	COMBINED MATHEMATICS	0.630	1,810
BIOLOGICAL SCIENCE	PHYSICS	0.629	23,895



The highest result among the filtered subject groups is **BUSINESS STATISTICS** in the **ARTS** stream, with an average Z-score of **0.9454** and 215 students.

10. Validate and Test the OLAP Cube

Several validation tests were performed to confirm that the OLAP cube accurately reflects the underlying data warehouse and that the analytical operations produce valid results.

Validation Test	Result	Status
Fact rows = Cube rows	1,012,659 = 1,012,659	PASS
Distinct students	337,553	PASS
Subjects	62	PASS
Streams	8	PASS
Syllabuses	2	PASS
Exam years	1	PASS
Duplicate fact records	0	PASS
Missing Student Keys	0	PASS
Missing Subject Names	0	PASS
Missing Stream Names	0	PASS
Missing Syllabus Names	0	PASS

Missing Source Values

Column	Missing Values
Z-score	315,747
Gender	4,452
District Rank	459,762
Island Rank	459,762

The missing values above originate from the source dataset. They do not indicate broken dimension joins. All important dimension keys and descriptive dimension names used by the cube contain zero missing values.

The duplicate fact-record test returned **0** duplicates, confirming that no duplicate student-subject fact records were detected.

11. Overall OLAP Validation Summary

Requirement	Status
Business questions defined	PASS
Data source identified	PASS
Dimensions defined	PASS
Hierarchies defined	PASS
Facts and measures selected	PASS
OLAP cube constructed	PASS
Roll-up implemented	PASS
Drill-down implemented	PASS
Slice implemented	PASS
Dice implemented	PASS
Charts generated	PASS
Cube row-count validation	PASS
Duplicate validation	PASS
Measure validation	PASS

The six required OLAP development steps have therefore been completed successfully.

12. Key Findings

- The Physical Science stream recorded the highest average Z-score among the stream categories with a value of 0.6380.
- The new syllabus recorded an average Z-score of 0.3072, compared with 0.2805 for the old syllabus.
- Female students recorded an average Z-score of 0.3729, while male students recorded 0.1952.
- The subject-level analysis identified BUSINESS STATISTICS within the ARTS stream as the highest average-Z-score subject among groups containing at least 100 students.
- The OLAP cube supports analysis from overall stream-level performance down to individual subject-level performance.
- The cube also supports multidimensional comparisons involving stream, gender and syllabus.

13. Conclusion

A complete student performance Data Warehouse and OLAP solution was successfully developed using PostgreSQL, Python, Pandas and Google Colab. The data was transformed into a star schema consisting of a central fact_student_subject table and five dimension tables.

The resulting OLAP cube enables multidimensional analysis of student performance using Z-score, rankings, student counts and grade distributions. Roll-up, drill-down, slice and dice operations were implemented successfully.

Validation confirmed that the cube contains the expected number of records and dimensions, contains no duplicate fact records, and maintains valid dimension relationships. The analytical results and visualizations demonstrate that the cube meets the defined business requirements.

Final Status: OLAP CUBE SUCCESSFULLY BUILT, ANALYSED AND VALIDATED